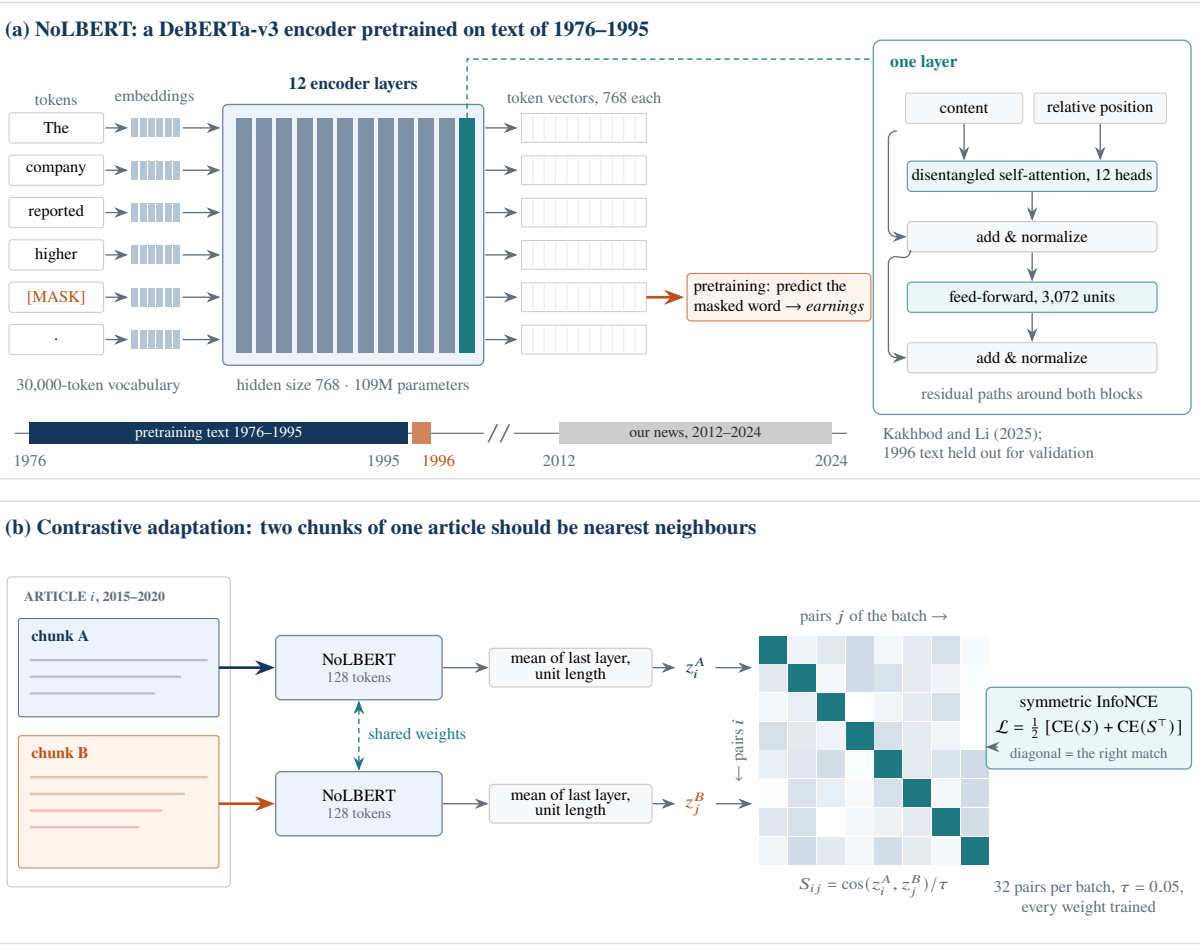

(a) NoLBERT: a DeBERTa-v3 encoder pretrained on text of 1976–1995
tokens
embeddings
12 encoder layers
token vectors, 768 each
The
company
reported
higher
[MASK]
.
30,000-token vocabulary
hidden size 768 · 109M parameters
pretraining: predict the masked word → earnings
one layer
content
relative position
disentangled self-attention, 12 heads
add & normalize
feed-forward, 3,072 units
add & normalize
residual paths around both blocks
pretraining text 1976–1995
our news, 2012–2024
1976
1995
1996
2012
2024
Kakhbod and Li (2025);
1996 text held out for validation
(b) Contrastive adaptation: two chunks of one article should be nearest neighbours
ARTICLE i, 2015–2020
chunk A
chunk B
NoLBERT
128 tokens
shared weights
NoLBERT
128 tokens
mean of last layer, unit length
mean of last layer, unit length
z_i^A
z_j^B
pairs j of the batch →
← pairs i
symmetric InfoNCE
$\mathcal{L} = \frac{1}{2}\,[\mathrm{CE}(S) + \mathrm{CE}(S^\top)]$
diagonal = the right match
$S_{ij} = \cos(z_i^A, z_j^B)/\tau$
32 pairs per batch, $\tau = 0.05$,
every weight trained